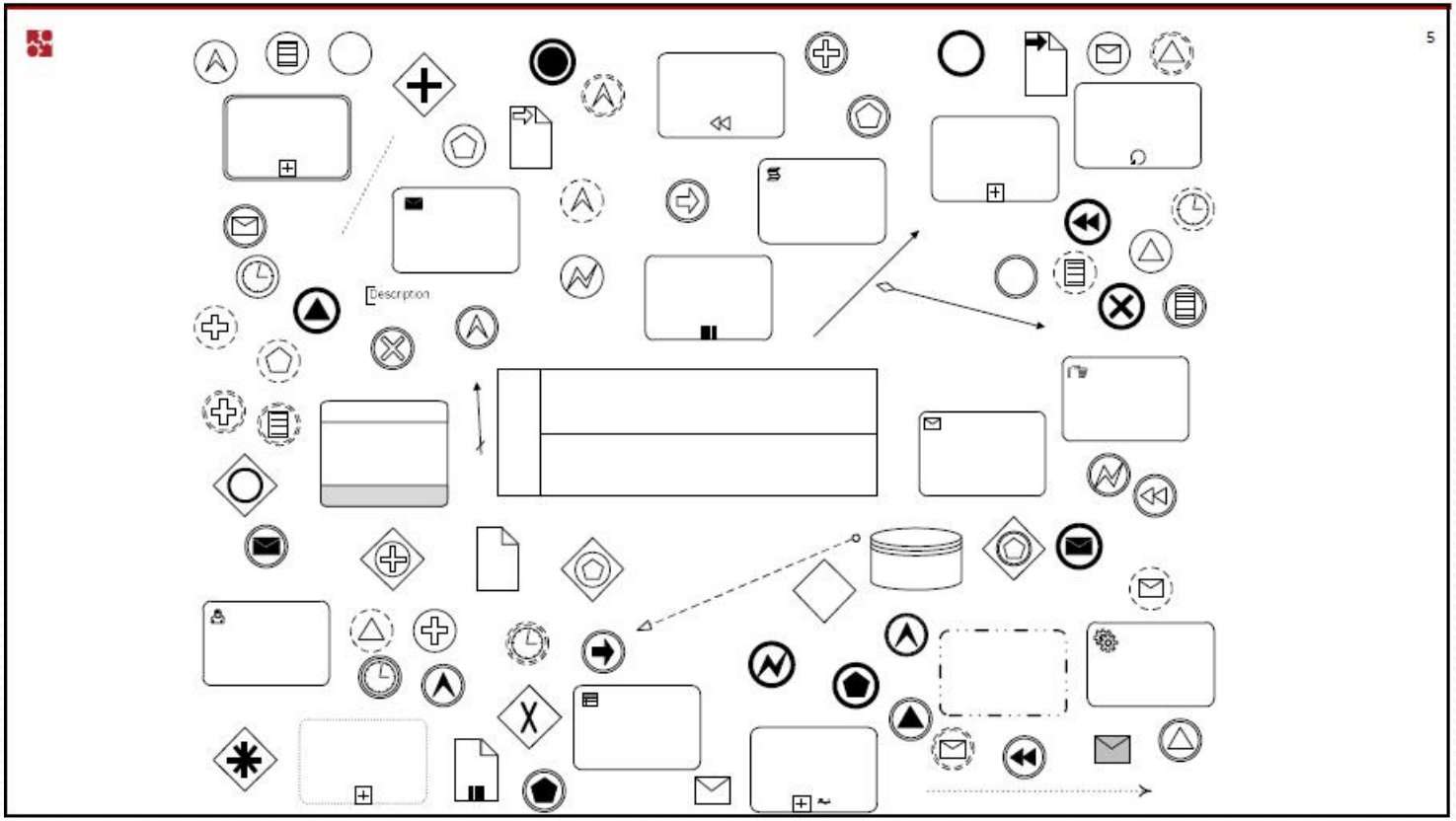
Process Modeling with BPMN 2.0

Business Process Model and Notation (BPMN)

- BPMN is a worldwide OMG standard
- Latest version is BPMN 2.0
- Lots of companies joined the standardization committee
- All major BPMS vendors go for BPMN

Business Process Model and Notation (BPMN)



Why should I care about BPMN?

- Business Process Model and Notation (BPMN) is the global standard for process modeling and one of the most important components of successful Business-IT-Alignment.
- More and more organizations are using BPMN and in more and more universities BPMN is taught as a subject.

Standard

- BPMN is not owned by a certain enterprise but by an institution (**OMG**), which is already established through other world-wide standards, e.g., UML.
- The standard is supported by many software products; you are less dependent on any particular vendor's products.

Simplicity

- The principle behind BPMN is rather simple which is why you can start working with this notation very quickly.

Why should I care about BPMN?

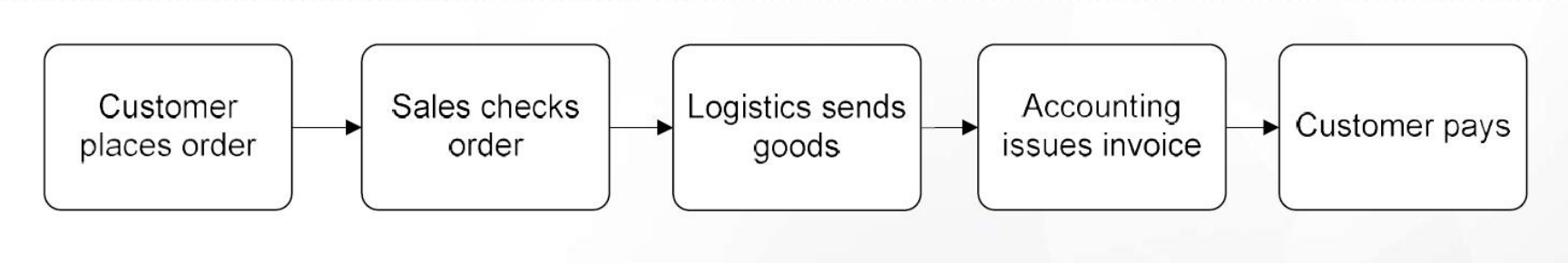
Power of expression

- If necessary, you can describe precisely how a process functions with BPMN.
- However, this is more difficult than only roughly describing the process.
- This way of precise modeling is possible, but not mandatory.

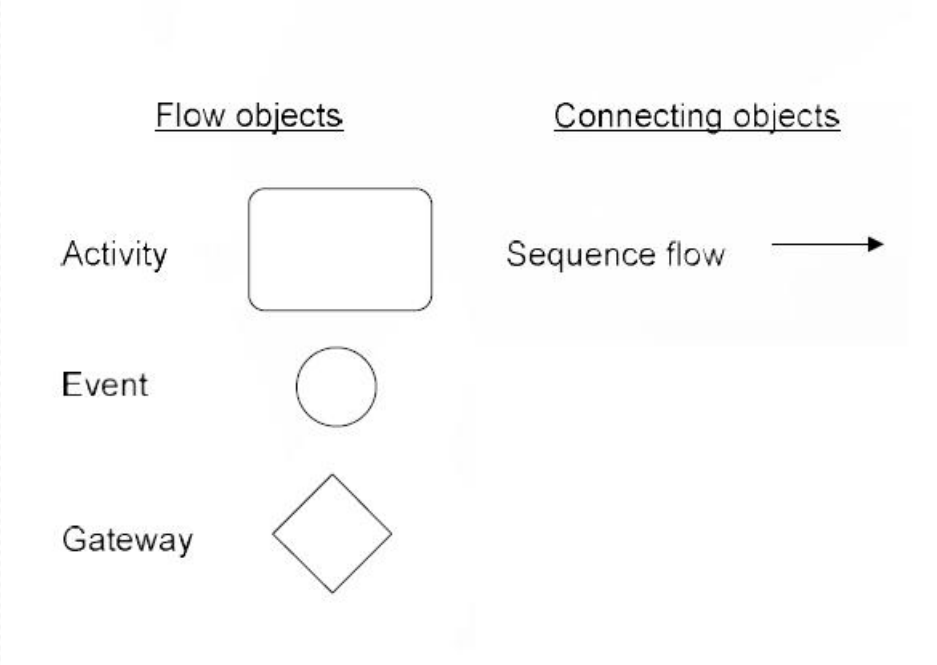
Implementation in IT

- BPMN has been primarily developed to support technical implementation of processes ("Process Automation").
- The more important the IT is in a company, the more helpful the use of BPMN becomes.

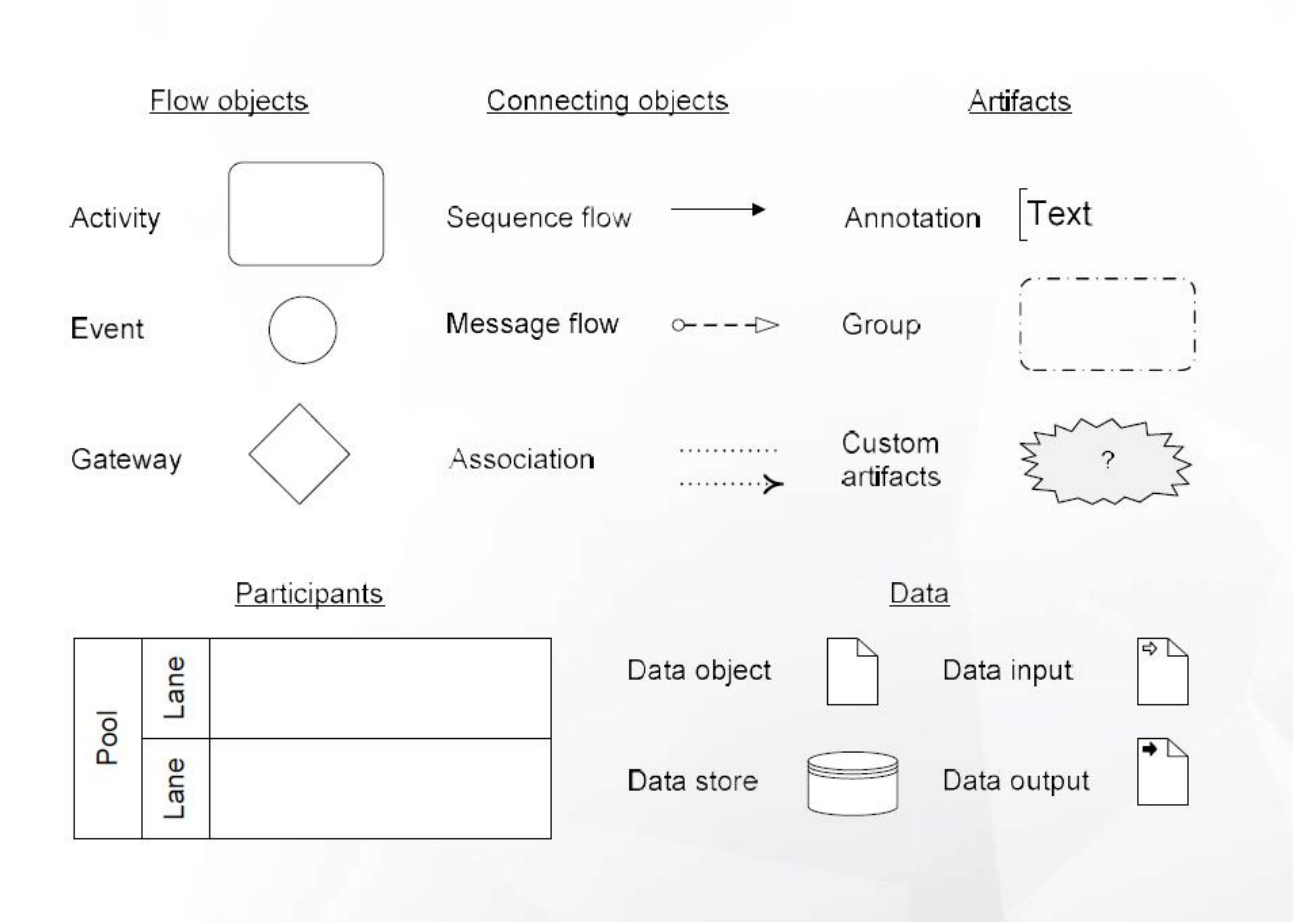
Our First BPMN Process Model



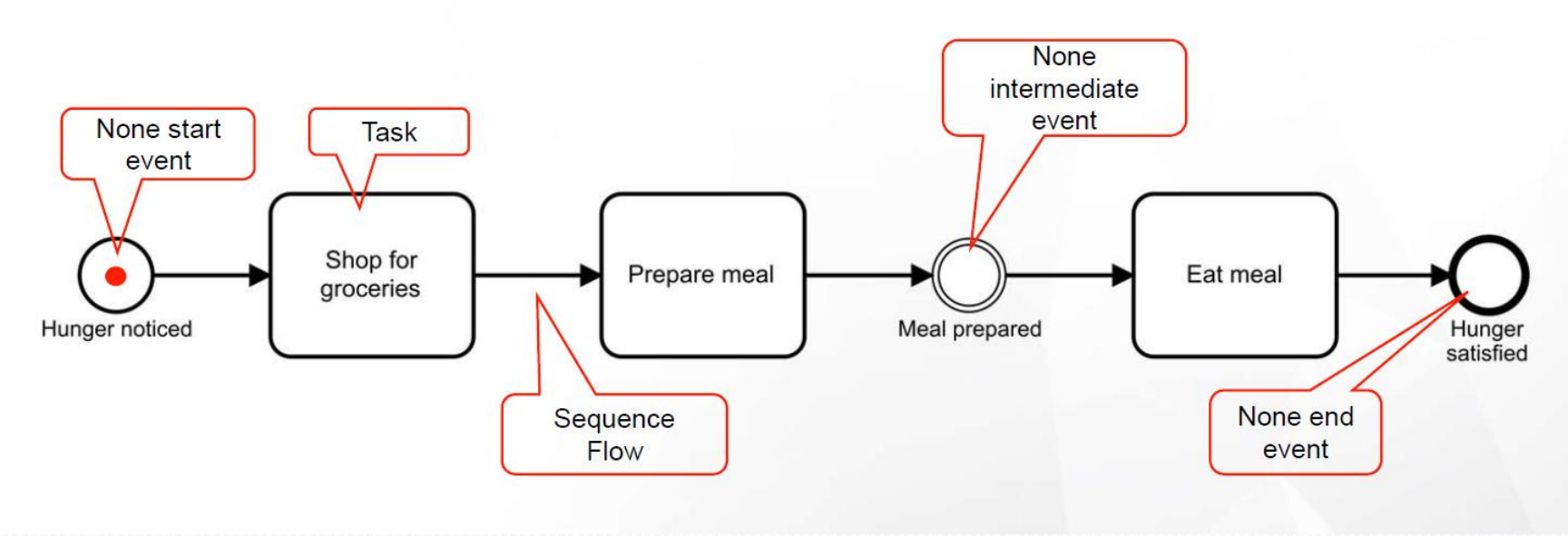
Basic Elements



Basic Elements - Documentation



Tasks and events



Tasks and events

Start Event

- Start events show which event causes the process to start.
- Start events are always catching events.

Task

- Tasks are the heart of the process.
- Ultimately, something has to happen to bring about a desired outcome.
- In BPMN, a task is technically part of the activities category, which also includes the sub-process.

Tasks and events

Intermediate Event

- Intermediate events represent a status that is reached in the process and that is modeled explicitly.
- They are used infrequently, but intermediate events can be useful, for example if you regard reaching a certain status as a milestone and you want to measure the time until that milestone is reached.
- Intermediate none events are always throwing events.

End Event

- End events mark the status reached at the end of the process path.
- End events are always throwing events.

Tasks and events

Best Practice: Naming Conventions

- When naming tasks, we try to adhere to the object-orientated design principle of using the [verb] + [object] pattern.
- We would say “acquire groceries,” or “Shop for groceries,” for example, not “first take care of shopping for groceries.”
- Events refer to something that has already happened regardless of the process (if they are catching events) or as a result of the process (if they are throwing events).
- For this reason, we use the [object] and make the [verb] passive in voice, so we write “hunger noticed.”
- BPMN does not require you to model start and end events for a process – you can leave them out – but **if** you model a start event, you must model an end event for each path.

Tasks and events

Best Practice: Naming Conventions

- The same is true for end events, which require start events.
- We always create our models with start and end events for two reasons: first, that way it's possible to determine the process trigger, and second, you can describe the final status of each path end.
- We only sometimes abandon this practice with sub-processes.

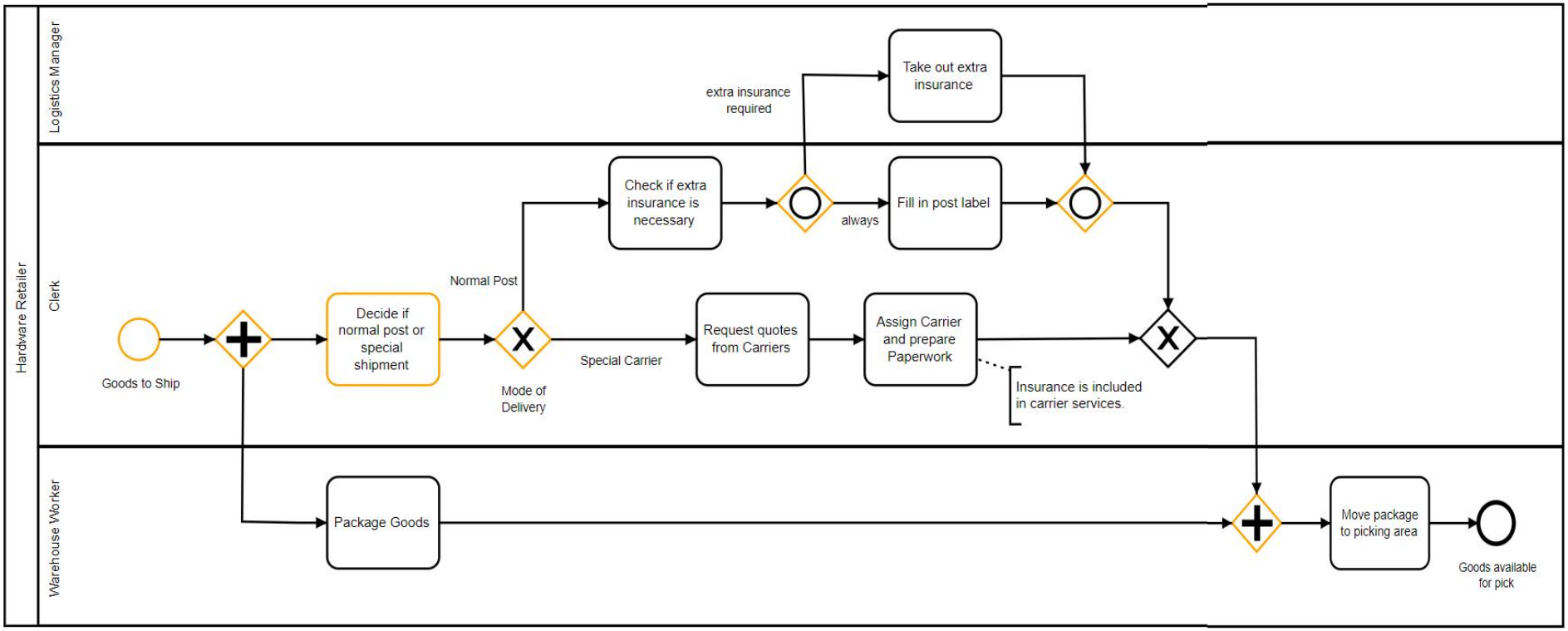
FAQ: Is it obligatory to draw BPMN diagrams horizontally? What if I prefer to draw them vertically?

- You could always draw your diagrams from top to bottom instead from left to right – the BPMN 2.0 standard does not forbid it.
- However, we do not recommend it: It is very uncommon, and experience has proven that people tend to understand the process flow better if it is described in the same way as written text (from left to right, at least in the western world).

Examples

Shipment Process

- In this diagram you can find the preparing steps a hardware retailer has to fulfill before the ordered goods can actually be shipped to the



Examples

Shipment Process

- Start Event
  - The plain start event ‘goods to ship’ indicates that this preparation should be done now.

- Parallel Gateway
  - Right after the instantiation of the process, there are two things done in parallel, as the parallel gateway indicates: While the clerk has to decide whether this is a normal postal or a special shipment(we do not define the criteria how to decide this inside the process model), the warehouse worker can already start packaging the goods.

Examples

Shipment Process

- Task
  - The clerk's task, which is followed by the exclusive gateway 'mode of delivery', is a good example of clarifying the recommended usage of an exclusive data based gateway:
  - The gateway is not responsible for the decision whether this is a special or a postal shipment.
  - Instead, this decision is undertaken in the activity before.
  - The gateway only work as a router, which is based on the result of the previous task, and provides alternate paths.
  - A task represents an actual unit of work, while a gateway is only routing the sequence flow.

Examples

Shipment Process

- Exclusive Gateway
  - This gateway is called ‘exclusive’ because only one of the following two branches can be traversed.
  - If we need a special shipment, the clerk requests quotes from different carriers, then assigns a carrier and prepares the paperwork.
  - However, if normal post shipment is fine, the clerk needs to check if extra insurance is necessary.

- Inclusive Gateway
  - If extra insurance is required the logistics manager has to take out that insurance.
  - In any case, the clerk to fill in a postal label for the shipment.

Examples

Shipment Process

- Inclusive Gateway
  - For this scenario, the shown inclusive gateway is helpful, because we can show that one branch is always taken, while the other one only if the extra insurance is required, but if it is taken, this can happen in parallel to the first branch.
  - Because of this parallelism, we need the synchronizing inclusive gateway right behind 'Fill in a Post label' and 'Take out extra insurance'.

- Inclusive Gateway(synchronizing)
  - In this scenario, the inclusive gateway will always wait for 'Fill in a Post label' to be completed, because that is always started.
  - If extra insurance is required, the inclusive gateway will also wait for 'take out extra insurance' to be finished.

Examples

Shipment Process

- Parallel Gateway(synchronizing)
  - Furthermore, we also need the synchronizing parallel gateway before the last task ‘add paperwork and move package to pick area’, because we want to make sure that everything has been fulfilled before the last task is executed.

Examples

Shipment Process

- In this example, we only used one pool and different lanes for the people involved in this process, which automatically means that we blank out the communication between those people:
  - We just assume that they are somehow communicating with each other.
  - If we had a process engine driving this process, that engine would assign user tasks and therefore be responsible for the communication between those people.
  - If we do not have such a process engine, but want to model the communication between the people involved explicitly, we would have to use a collaboration diagram. We will see this in next example.

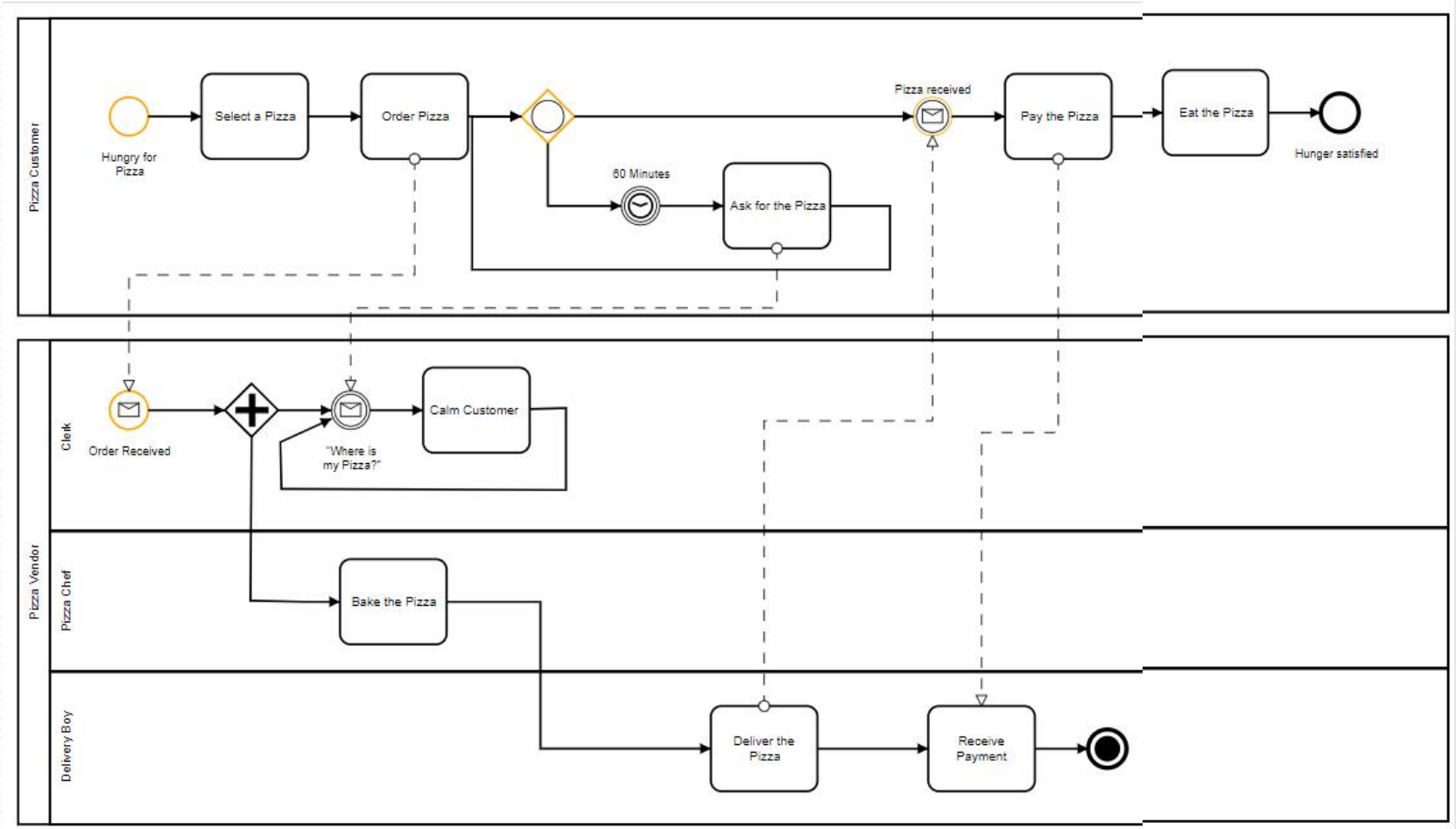
Examples

Pizza Collaboration

- This example is about Business-To-Business-Collaboration.
- Because we want to explicitly model the interaction between a pizza customer and the vendor, we have classified them as “participants”, therefore providing them with dedicated pools:

Examples

Pizza Collaboration



Examples

Pizza Collaboration

- Start Event
  - If we step through the diagram we should start with the pizza customer, who has noticed her stomach growling.
  - The customer therefore selects a pizza and orders it.

  - Let's have a closer look at the vendor process now.
  - It is triggered by the order of the customer, as shown by the message start event and the message flow going from 'order a pizza' to that event.
  - After backing the pizza, the delivery boy will deliver the pizza and receive payment.

Examples

Pizza Collaboration

- Event based Gateway
  - The customer waits for the pizza to be delivered.
  - The event based gateway indicates that the customer actually waits for two different events that could happen next:
  - Either the pizza is delivered, as indicated with the following message event, or there is no delivery for 60 minutes, i.e., after one hour the customer skips waiting and calls the vendor, asking for the pizza.
  - We now assume that the clerk promises the pizza to be delivered soon, and the customer waits for the pizza again, asking again after the 60 minutes and so on.

Examples

Pizza Collaboration

- Message Event
  - In this example, we use message events not only for informational objects such as the pizza order, but also for physical objects, like the pizza.
  - We can do this because those physical objects actually inherently act as informational objects:
  - When the pizza arrives at the customer's door, she will recognize this arrival and therefore know that the pizza has arrived, which is exactly the purpose of the accordant message event in the customer's pool.
  - Of course we can only use the model that why because this example is not meant to be executed by a process engine.

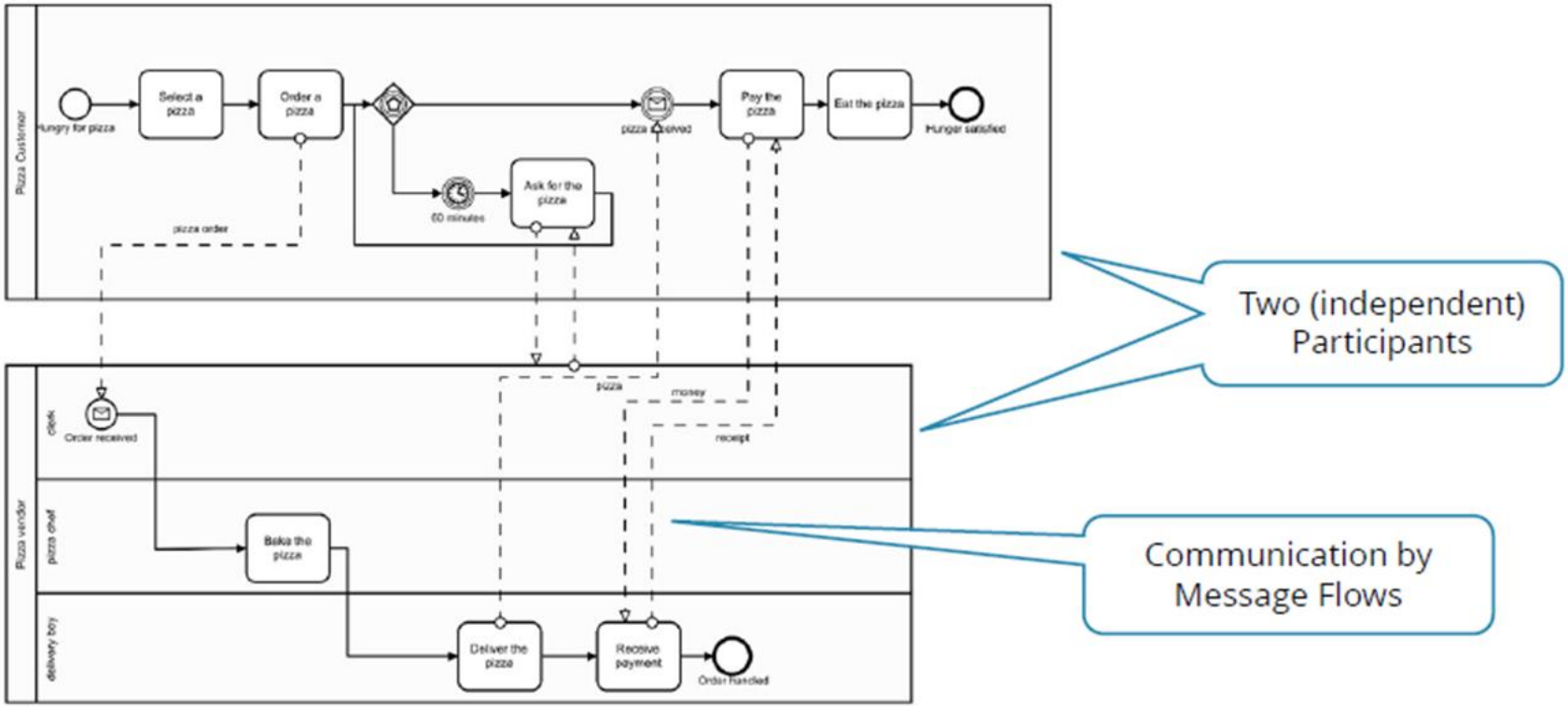
Examples

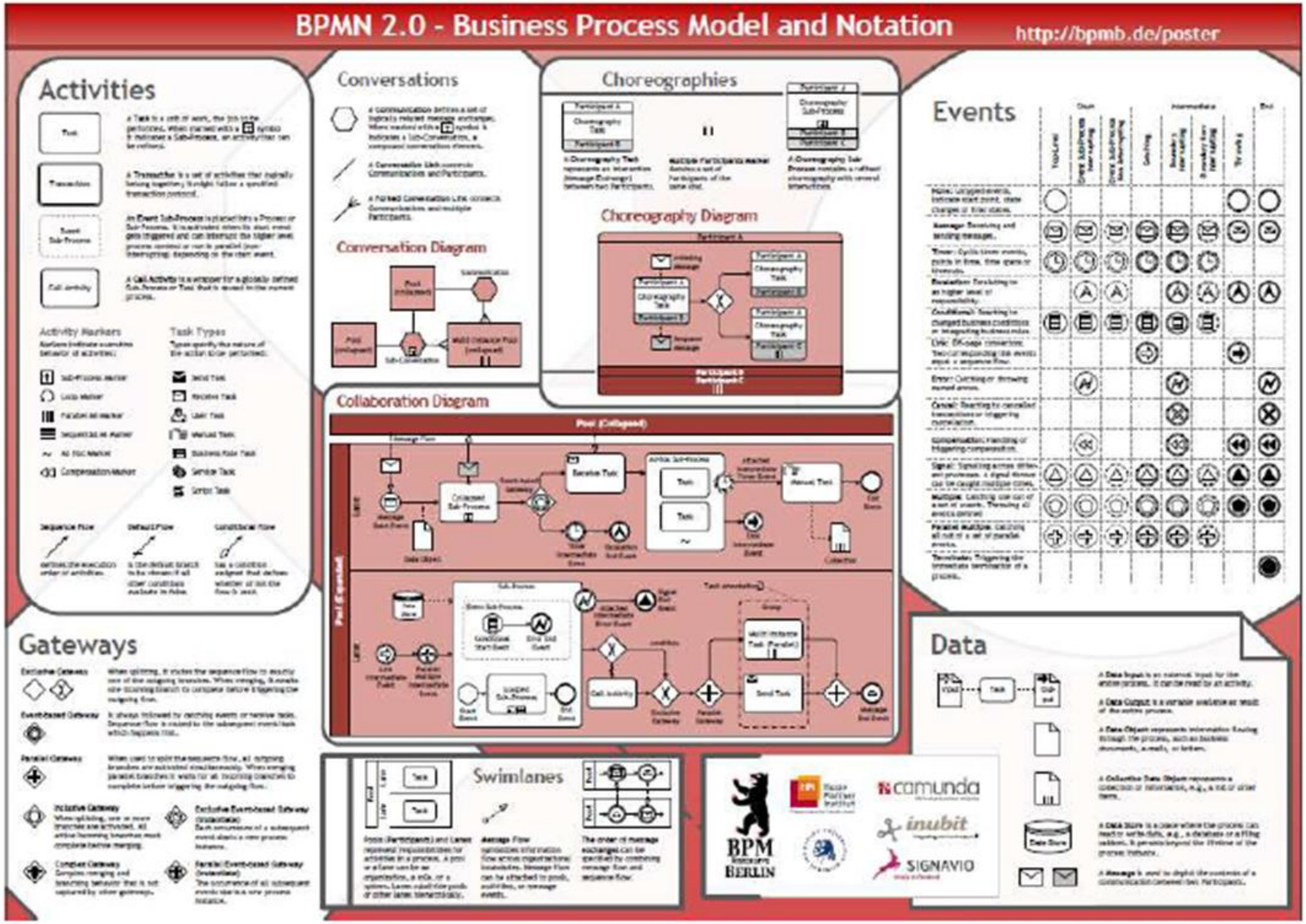
Pizza Collaboration

- Please note that there are no default semantics in this type of modeling, which means you can model collaboration diagrams to show the interaction between business partners, but also zoom into one company, modeling the interaction between different departments, teams or even single workers and software systems in collaboration diagrams.
- It is totally up to the purpose of the model and therefore a decision the modeler has to make, whether a collaboration diagram with different pools is useful, or whether one should stick to one pool with different lanes, as shown in the previous example.

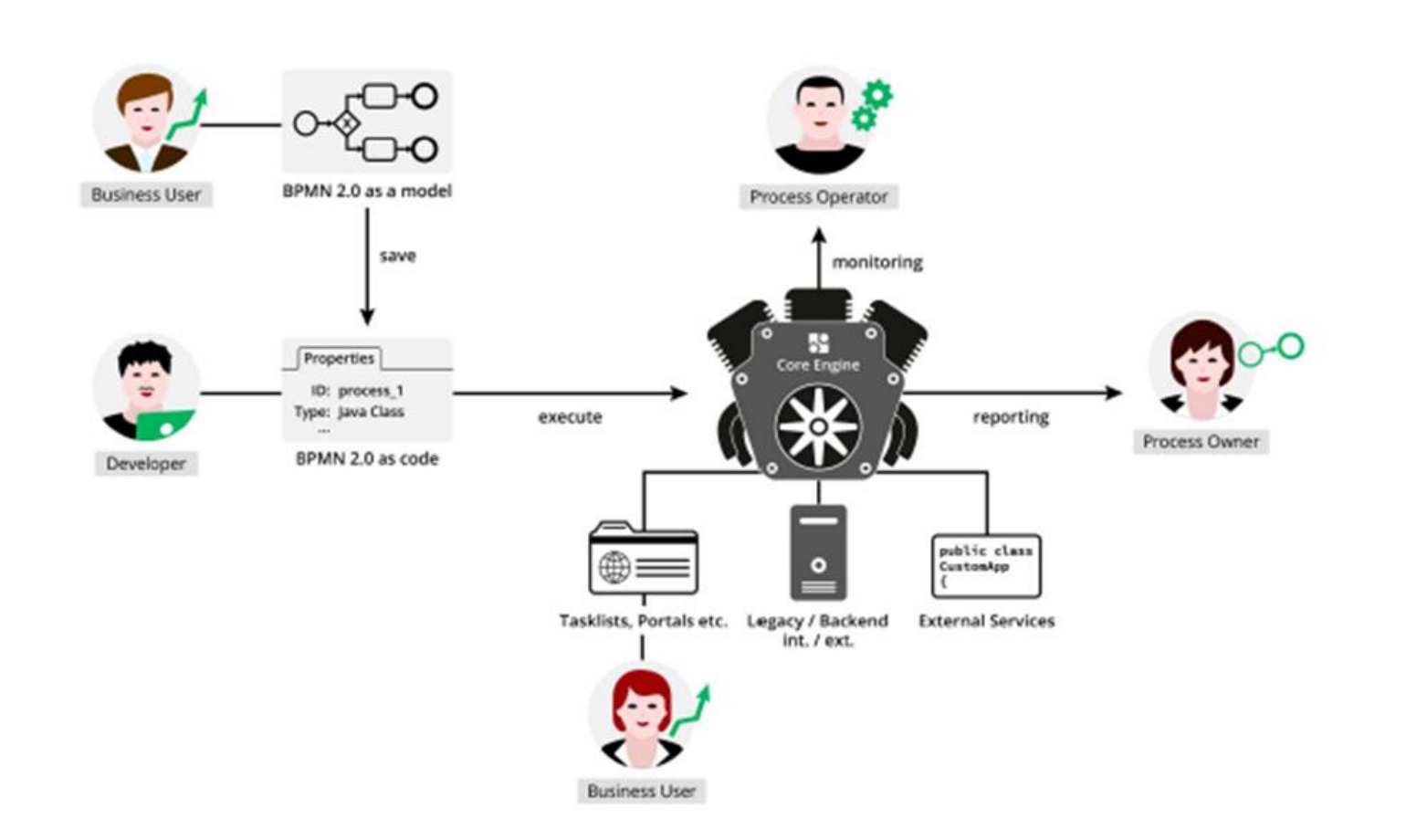
Examples

Pizza Collaboration

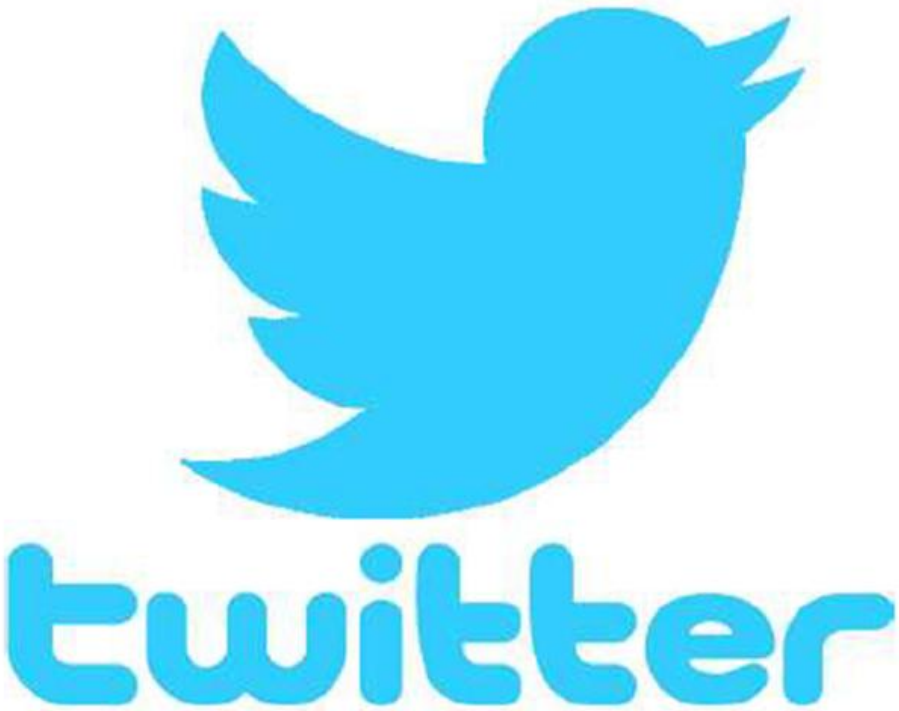



BPMN 2.0 - Business Process Model and Notation
http://bpmb.de/poster
Activities
Activity Markers
Task Types
Conversations
Conversation Diagram
Choreographies
Choreography Diagram
Events
Collaboration Diagram
Gateways
Swimlanes
Data
BPM Berlin
comunda
inubit
SIGNAVIO

What’s So cool about BPMN 2.0



You know Twitter?



Imagine the Anti Agile Inc.

Why do we need Twitter?

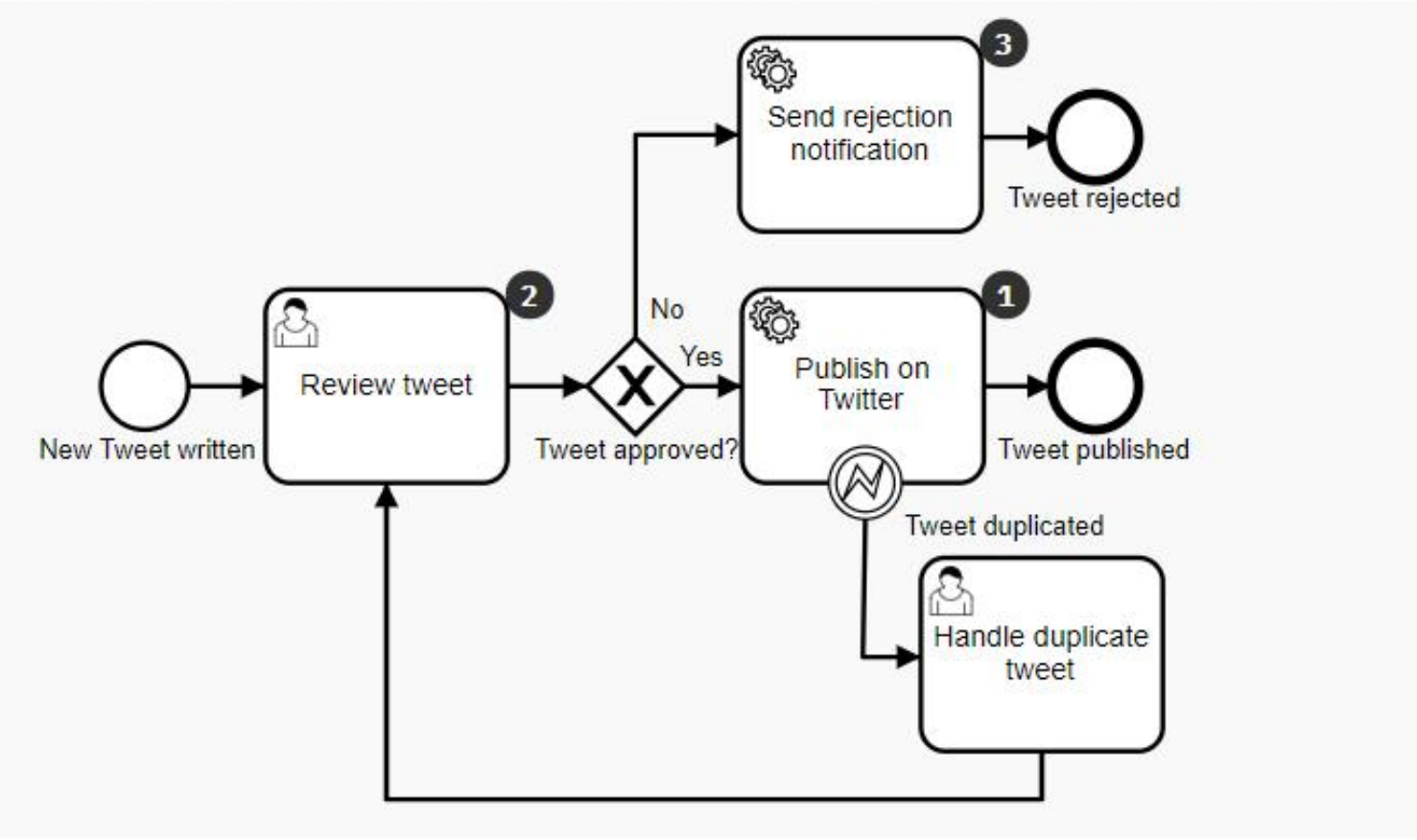
- We do not trust our employees.
- We want to control everything.
- We do not like Twitter,
- But we want to be part of it.

Group exercise 0

Draw a Twitter QA Process Model

- Goal
  - Discuss as a group a common process model to fulfill the needs of Anti Agile Inc.
- Short description
  - Try to find the activities and events of the process to fulfill the needs of Anti Agile Inc.
  - Model the process on a whiteboard or flipchart.
- Detailed Steps
  - Discuss the requirements of Anti Agile Inc.
  - Model the Twitter QA process as a group on a whiteboard or flipchart.
  - Discuss the process model(s) with the trainer afterwards.

Group exercise 0



Workflow Automation Use Cases

Human Task Management

System Integration